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Elements de la Théorie de l'information
Problem Set 8

Problem 1 $\quad X \to Y \to Z$

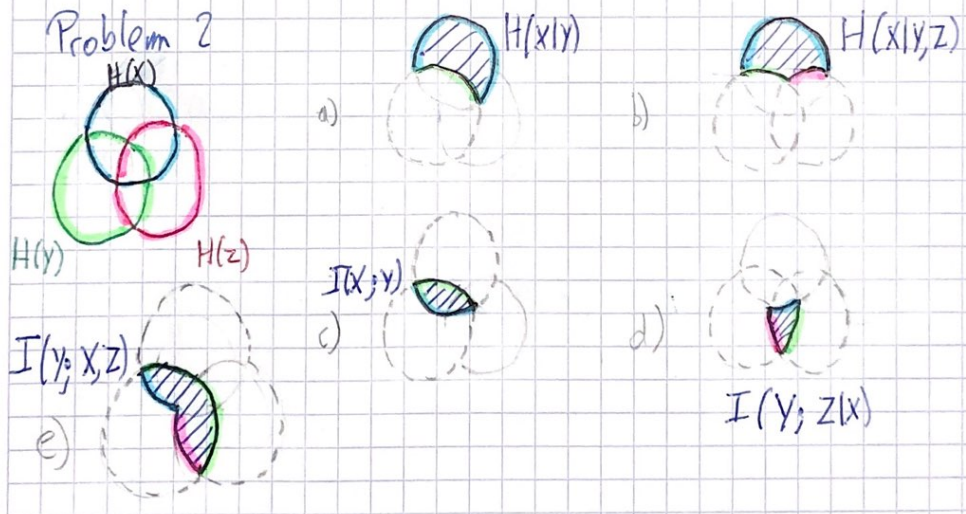
$$I(X;Y,Z) = I(X;Z) + I(X;Y|Z)$$
$$= I(X;Y) + I(X;Z|Y)$$

$$I(X;Z|Y) = H(X) + H(Z|Y) - H(X,Z|Y) = H(X) + H(Z|Y) - H(X) - H(Z|Y,X)$$
$$\overset{X \perp Z}{=} H(Z|Y) - H(Z|Y) = 0$$

$$\Rightarrow \quad I(X;Y,Z) = I(X;Y) = I(X;Z) + I(X;Y|Z)$$

$$(\forall \dot{X} \; I(\dot{X}) \geq 0) \Rightarrow \begin{cases} I(X;Y) \geq I(X;Z) \\ I(X;Y) \geq I(X;Y|Z) \end{cases}$$

Problem 2



Problem 3 $\quad X \to Y \to Z$

a) $I(X;Z) \overset{\text{problem 1}}{\leq} I(X;Y) \overset{\text{mutual info decomposes via entropy}}{=} H(Y) - \underbrace{H(Y|X)}_{\leq 0} \leq H(Y) \overset{\text{uniform pdf has the largest entropy}}{\leq} \log_2|\mathcal{Y}|$

$$I(X;Z) \leq \log_2(k)$$

b) $k=1 \Rightarrow \log_2(k) = \log_2(2^0) = 0$

$(0 \leq)\ I(X;Z) \overset{a)}{\leq} \log_2(k) = 0$

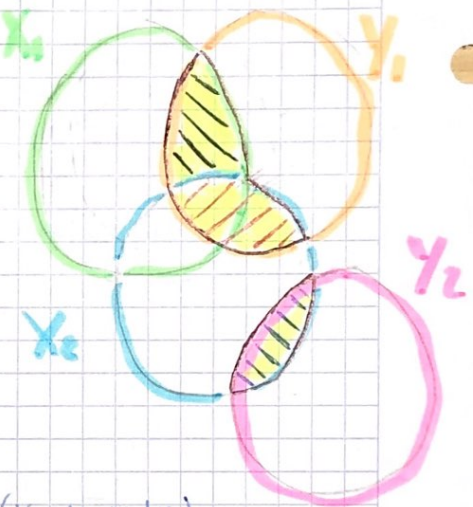
$I(X;Z)=0 \Leftrightarrow X \perp Z$

X and Z are independant

Problem 4

$$2H(X_2) + H(X_1|X_2) - H(X_2|Y_1) - H(X_2|Y_2) - H(X_1|X_2,Y_1) = \left[H(X_2) - H(X_2|Y_2)\right] + \left[H(X_2) - H(X_2|Y_1)\right] + \left[H(X_1|X_2) - H(X_1|X_2,Y_1)\right]$$

$$= I(X_2;Y_2) + I(X_2;Y_1) + I(X_1;Y_1|X_2)$$



$$\left.\begin{array}{l} X_1 \longrightarrow Y_1 \\ X_2 \nearrow \\ X_2 \longrightarrow Y_2 \end{array}\right\} \begin{array}{l} H(Y_2|X_1) = H(Y_2) \ (-I(X_1,Y_2)=0) \\ \Leftrightarrow \\ I(X_1,Y_2)=0 \end{array}$$

$$I(X_1,X_2;Y_1,Y_2) = I(X_2;Y_2) + I(X_2;Y_1) + I(X_1;Y_1|X_2)$$

$$\Rightarrow \boxed{I(Y_1,Y_2)=0} \ ?!$$

a) $I(X_1,X_2;Y_1,Y_2) = \underbrace{I(X_2;Y_1,Y_2)}_{(\Delta)} + \underbrace{I(X_1;Y_1,Y_2|X_2)}_{(\square)}$

$(\Delta)\ I(X_2;Y_1,Y_2) = I(X_2;Y_2) + I(X_2;Y_1|Y_2)$

$(\square)\ I(X_1;Y_1,Y_2|X_2) = I(X_1;Y_1|X_2) + \underbrace{I(X_1;Y_2|X_2,Y_1)}_{I(X_1;Y_2)=0}$

$$I(X_1,X_2;Y_1,Y_2) = \underbrace{\left[H(X_2) - H(X_2|Y_2)\right] + \left[H(X_2|Y_2) - H(X_2|Y_2,Y_1)\right]}_{(\Delta)} + \underbrace{\left[H(X_1|X_2) - H(X_1|X_2,Y_1)\right]}_{(\square)} + 0$$

$$= H(X_2) - H(X_2|Y_2,Y_1) + H(X_1|X_2) - H(X_1|X_2,Y_1)$$

b) $X_1 \perp X_2 \Rightarrow \begin{cases} H(X_1|X_2) = H(X_1) \ \boxed{(-I(X_1,X_2)=0)} \\ H(X_1|X_2,Y_1) = H(X_1,Y_1) \end{cases}$

$$\Rightarrow I(X_1,X_2;Y_1,Y_2) = H(X_2) - H(X_2|Y_2,Y_1) + \underbrace{\left[H(X_1) - H(X_1|Y_1)\right]}_{=\emptyset}$$

Problem 4

With $\underline{Y_1 \perp Y_2}$: a) $I(X_1, X_2; Y_1, Y_2) = I(X_2; Y_1, Y_2) + I(X_1; Y_1, Y_2 | X_2)$

$= \left[I(X_2; Y_2) + I(X_2; Y_1 | Y_2) \right] + \left[I(X_1; Y_1 | X_2) + I(X_1; Y_2 | X_2, Y_1) \right]$

$= I(X_2; Y_2) + I(X_2; Y_1) + I(X_1; Y_1 | X_2) + 0 \quad \{ 0 \geq I(X_1; Y_2) \geq I(X_1; Y_2 | X_2, Y_1) \geq 0 \}$

$= \left[H(X_2) - H(X_2 | Y_2) \right] + \left[H(X_2) - H(X_2 | Y_1) \right] + \left[H(X_1 | X_2) - H(X_1 | X_2, Y_1) \right]$

$= 2H(X_2) + H(X_1 | X_2) - H(X_2 | Y_1) - H(X_2 | Y_2) - H(X_1 | X_2, Y_1)$ ⌋

b) $X_1 \perp X_2 \Rightarrow \begin{cases} H(X_1 | X_2) = H(X_1) \\ H(X_1 | X_2, Y_1) = H(X_1 | Y_1) \end{cases}$

$I(X_1, X_2; Y_1, Y_2) = 2H(X_2) + H(X_1) - H(X_2 | Y_1) - H(X_2 | Y_2) - H(X_1 | Y_1)$ ⌋